

Hardware Handout

Three reference sheets for building and debugging this station's boards — pulled from real firmware and real hardware sessions, not generic datasheet redraws. Print double-sided; each section starts on its own page.

1. Board & sensor pinouts — every board wired into the synth
2. DHT22 / DS18B20 sensor wiring — the shared D5 pull-up circuit
3. OLED status icons — what the synth board's display is telling you

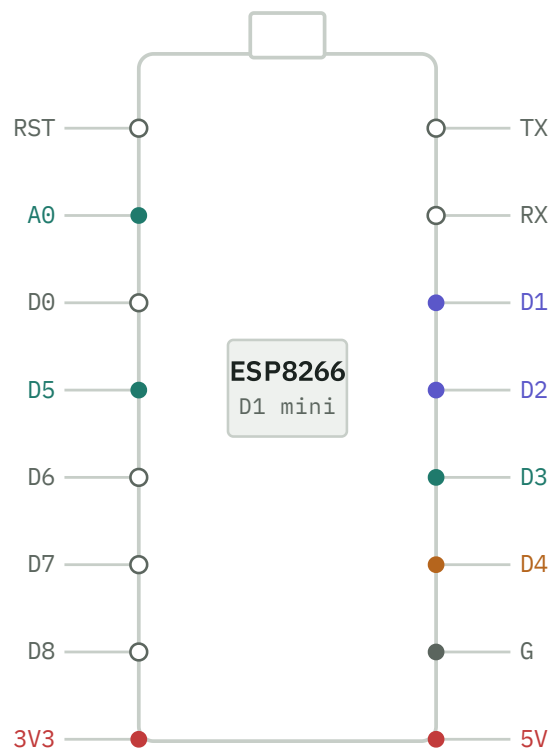
STATION 5 — HARDWARE REFERENCE

Noizetoyz Pinouts

Real pin assignments for every board wired into the synth, pulled straight from this repo's firmware — not a generic datasheet redraw. **Copper** traces the actual audio signal across boards: GPIO out on the Wemos or UNO, in and back out the far side of the LM386.

Wemos D1 mini

ESP8266 · esp_synth / bmp180 / note_player / dht22



D1/D2 pull double duty — I²C bus for the sensor and OLED sketches, plain buttons in the synth sketch. Same physical pins, different job per sketch.

`esp_synth.ino` — D1/D2/D3 buttons, A0 pot (fx amount), D4 fixed Mozzi audio out.

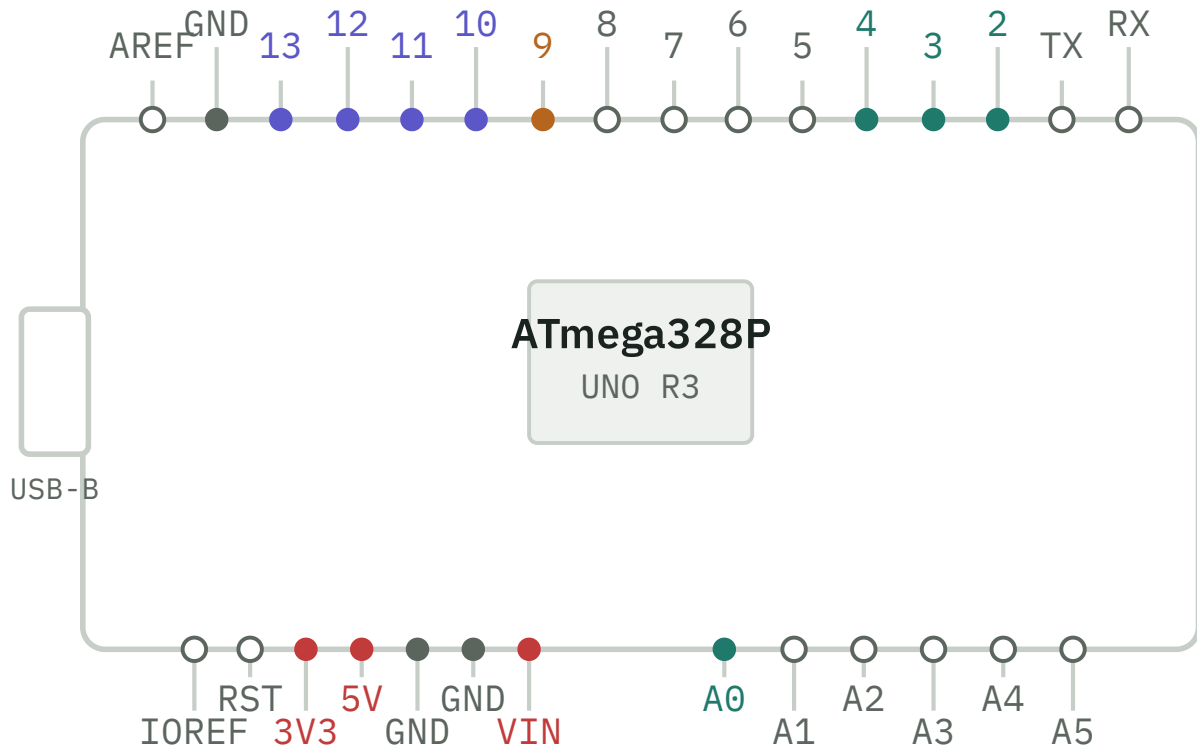
`bmp180_wifi.ino` / `bmp180_smoke_test.ino` — D2 SDA, D1 SCL.

`esp_note_player.ino` — D4 audio out, D1/D2 I²C to the OLED shield.

`dht22_wifi.ino` / `ds18b20_test.ino` — D5 sensor data line (a separate board — see the wiring section).

Arduino UNO

ATmega328P · uno_synth / uno_ethernet



`uno_synth.ino` — pins 2/3/4 buttons, A0 pot, pin 9 Mozzi audio out (per Mozzi's own pin table).

`uno_ethernet_smoke_test.ino` — W5100-class shield on the standard SPI bus (11/12/13), CS on pin 10.

LM386 amp module

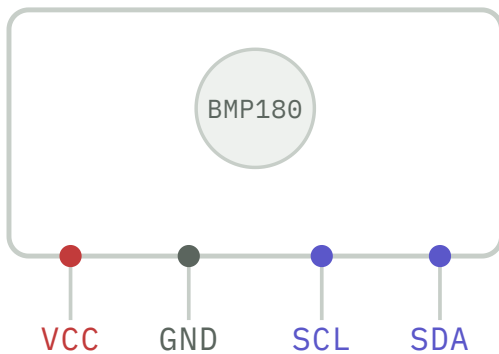
GS26347 · ×4 channels on one strip



IN takes the filtered PWM straight off the Wemos/UNO's audio pin; **OUT** drives a speaker. One strip carries four identical channels — enough for every hardware variant in the room at once.

BMP180

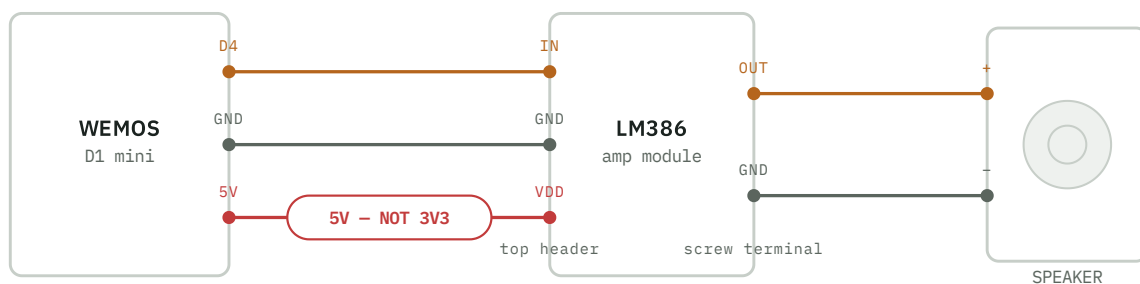
temp / pressure · I²C, addr 0x77



Same four pins regardless of which relay it feeds — station-2's `wifi_sensor_relay.py` or station-5's `bmp180_wifi.ino`, both read it over I²C. Confirmed publishing live readings end-to-end 2026-09-04.

Wiring: Wemos → amp → speaker

signal chain · 3 connections, 1 gotcha



Three wires, no more: **D4** + a **GND** off the Wemos into the amp module's top header **IN** + **GND**; the amp's bottom screw terminal **OUT** + **GND** straight to a speaker. The one gotcha is power — the amp's **VDD** comes off the Wemos's **5V** pin, **not 3V3**. The regulated 3.3V rail only has ~200–300mA of headroom and is already feeding the ESP8266's own WiFi radio; the LM386 wants upward of 4V to behave, and sharing that rail risks brownouts or resets under load.

● power ● ground ● signal (button / pot / data) ● bus (I²C / SPI)

● audio path ○ unused here

Traced from this repo's firmware (`station-5-synth/firmware/`, `station-2-live-data/wifi_sensor_relay.py`) and real hardware photographed 2026-09-01 through 2026-09-04 — pin roles reflect what these sketches actually do.

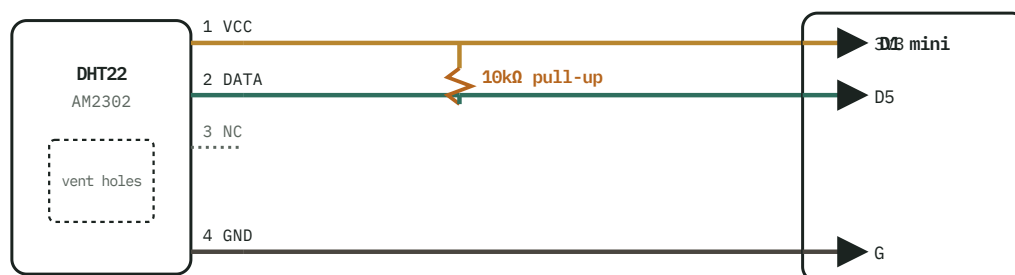
DHT22 / DS18B20 sensor wiring

Both sensors share one circuit: D5 for data, 3V3 for power, a pull-up bridging the two — confirmed 2026-09-04 by swapping sensors on the exact same untouched wiring (this is exactly how a dead DHT22 unit was isolated from an otherwise healthy circuit). Only the physical pin order differs.

— power — data + pull-up — ground

DHT22 / AM2302

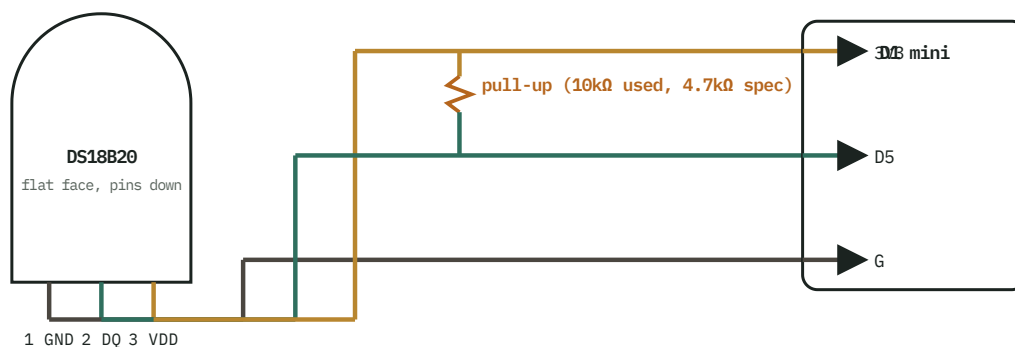
bare sensor on breadboard



VCC→3V3, GND→G, DATA→D5. Pin 3 (NC) is left unconnected. The 10kΩ pull-up bridges the VCC rail and the DATA line right at the sensor end — bare sensors like this one carry no onboard pull-up, unlike a breakout PCB.

DS18B20

T0-92, flat face toward you, pins down



VDD→3V3, GND→G, DQ→D5 — same three D1 mini pins the DHT22 uses, same pull-up placement. Datasheet-spec pull-up is 4.7kΩ; the 10kΩ already on the breadboard from the DHT22 wiring works fine left in place.

WHY 3V3, NOT 5V

Both sensors are happy on 3.3–6V, but the ESP8266's GPIOs are not 5V-tolerant (~3.6V absolute max). 3V3 keeps DATA/DQ's HIGH level matched to the ESP8266's own logic level.

WHY D5 SPECIFICALLY

GPIO14 — one of the DHT library's own documented safe ESP8266 pins. D3/D4 (GPIO0/2) have boot-mode strapping behavior; D5 doesn't collide with I²C (D1/D2) or Mozzi's audio pin (D4).

WHY THE PULL-UP MATTERS

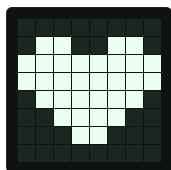
Both sensors use a single data line that must be pulled high when idle. No pull-up (or one bridging the wrong lines) reads as total silence — indistinguishable from a dead sensor without a cross-check.

What actually varies between them

Everything on the D1 mini side of both diagrams is identical — pins, pull-up placement, ground rail. All that changes is the sensor's own physical pin order. Swapping one sensor for the other is a drop-in test of the sensor alone — exactly how `firmware/esp-wifi/ds18b20_test/` isolated a dead DHT22 unit from an otherwise completely healthy circuit on 2026-09-04.

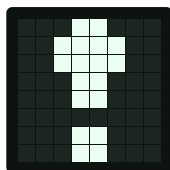
Noizetoys status icons

8×8, one bit per pixel — the real constraint of the SSD1306 shield wired to every board. These four are flashed and confirmed on real hardware, right-aligned in the footer row of the synth board's display.



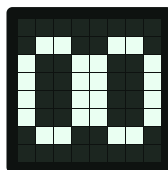
Heart

relay config fetched
from real
ATProto/HappyView



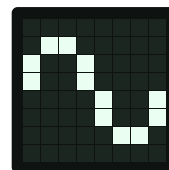
Exclamation

DEFAULT_RELAY_HOST
fallback in use



Link

downlink connected



Broken link

downlink not connected

Byte order matches PROGMEM arrays: one byte per row, MSB first, top to bottom. Right-aligned in the footer row, opaque-masked over the scrolling marquee text so it never shows through.